

# Lecture 9

$\neq Ax = 0$  for  $m > n$  (more unknowns than equations)

then non zero  $x$  would be solutions.

$\Rightarrow$  Linear Independence

Vectors  $v_1, v_2, \dots, v_n$  are linearly independent if no linear combination except all zeros give 0.

$$\Rightarrow a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$$

where all  $a_i \neq 0$  then vectors are linearly dependent.

# If any of these vectors is the zero vector then the vectors are definitely dependent.

eg  $V_1 = 0$

$$aV_1 + 0V_2 + 0V_3 \dots + 0V_n = \underline{0}$$

$$\forall a \in \mathbb{R} / \mathbb{C}$$

# If there is anything in the nullspace other than the 0 vector, then the vectors are dependent

$$Ax = 0$$

$\exists x \neq 0$

then there is a linear combination of  $C(A)$  for which it is 0

# Rank  $r$  of matrix is  $r = n$   
then the columns are independent.

$r < n \rightarrow$  dependence -

## Span

# The vectors  $v_1, \dots, v_n$  span a space  $S$  then  $S$  consists of all linear combinations of  $S$ .

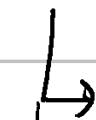
$\Rightarrow$  The smallest space which contains all the vectors  $v_1, \dots, v_n$  is the span of these vectors

Basis  $\rightarrow$  A basis for a vector space is a sequence of vectors  $v_1, \dots, v_d \rightarrow$

- They are independent.
- They span the vector space.

Eg

$\mathbb{R}^3$



$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\textcircled{I}$

$$N(I) = \{0\}$$



$$\begin{bmatrix} 1 & 1 & 4 \\ 2 & 1 & 9 \\ 3 & 1 & -5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 2 & 5 & 8 \end{bmatrix}$$

①

$$\textcircled{3} = 2 \times \textcircled{2} - \textcircled{1}$$

Now to check if the cols are independent we do elimination & find the rank of a matrix.

- For a square matrix  $n \times n$ , if  $\mathbb{R}^2$ , it must be invertible

Imp "dimension" of space = # of vectors

And all the basis for a given space have equal number of vectors.

- The number of vectors in the basis of a space is its dimension.

eg

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

- span is the  $C(A)$
- They are not a basis as they are dependent.
- $N(A)$  is not just zero

eg

$$\begin{bmatrix} 1 \\ 1 \\ -1 \\ 0 \end{bmatrix} \text{ is } \underline{\underline{\text{in } N(A)}}$$

- Basis for  $C(A) \rightarrow$  cols 1 & 2

- Basis of  $C(A)$  can be found by taking the pivot cols.

$$\underline{\text{rank}(A)} = \underline{\# \text{ pivot cols}} = \underline{\text{dimension}(C(A))}$$

$$\text{dimension}(N(A)) = [n - \text{rank}(A), 1]$$

for given ex  $\boxed{\underline{\text{dim}(N(A)) = 2}}$

$$\underline{n - \text{rank}(A)} = \underline{\# \text{ free vars}} = \underline{\text{dim}(N(A))}$$